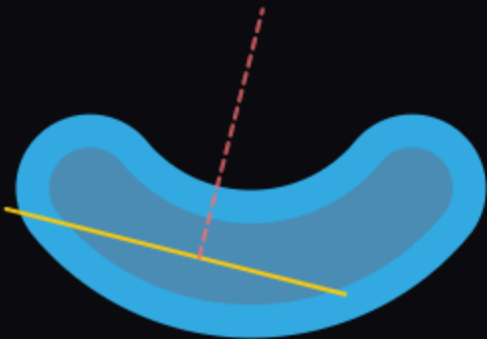


Fiber Bend & TIR Loss

Curved core: outer-wall incidence vs θ_c ; tight R leaks (meridional sketch).

Outer hit: $\theta_i \approx 89.7^\circ > \theta_c$ — ray leaks (schematic refraction)



Paraxial tilt added at outer wall: $\Delta\theta \approx a / R$ (normalized $a = 1$)

FIBER & BEND

n_{core} 1.465

n_{cladding} 1.435

Ray angle (axis) 0°

Bend radius R 220

Tighter bends tilt meridional rays at the **outer** cladding; when incidence exceeds **$\theta_c = \arcsin(n_{\text{clad}}/n_{\text{core}})$** , power leaks (macro-bending; micro-bends and mode coupling omitted).

SHORTCUTS

- Reduce bend radius until the outer hit loses TIR
- Compare θ_c to the sketched incidence at the outer wall

MEASURED VALUES

θ_c 78.38°

$\Delta\theta$ (a/R) 0.26°

About this model

Optical fibers guide light by total internal reflection (TIR) at the core-cladding interface, requiring the ray's angle of incidence to exceed the critical angle. This simulator visualizes how bending an optical fiber can cause light to leak out, a primary source of signal loss in real-world fiber optic systems. It focuses on meridional rays—those that pass through the fiber's central axis—traveling through a curved fiber segment. The core physics is governed by Snell's Law, which defines the critical angle $\theta_c = \arcsin(n_{\text{cladding}} / n_{\text{core}})$. In a straight fiber, a ray with an angle θ_i (measured from the axis) less than the acceptance angle will be trapped. However, when the fiber is bent with a radius of curvature R, the geometry of the ray's path relative to the outer wall changes. The ray's effective angle of incidence on the curved interface decreases. If the bend radius R is too small, this effective angle falls below θ_c , violating the TIR condition and causing the ray to refract out of the core. The simulator simplifies the complex 3D nature of light propagation by using a 2D cross-section and considering only meridional rays, ignoring skew rays and modal effects. It also assumes a perfect step-index fiber with sharp, smooth interfaces and neglects absorption and scattering losses. By adjusting parameters like bend radius, core/cladding refractive indices, and input ray angle, students can explore the relationship between bend tightness, incidence geometry, and TIR failure. They learn to predict the minimum safe bend radius for a given fiber and input condition, connecting fundamental optics principles to a critical engineering constraint in telecommunications and medical endoscopy.

Who it's for: Undergraduate students in optics, photonics, or electrical engineering courses covering waveguide theory and fiber optic fundamentals.

KEY TERMS

- Total Internal Reflection (TIR)
- Critical Angle
- Snell's Law
- Refractive Index
- Bend Loss
- Meridional Ray
- Step-Index Fiber
- Acceptance Angle

How it works

Complements **Numerical Aperture**: even rays inside the acceptance cone can **radiate** if the guide curves too sharply — the same **step-index** interface, but geometry breaks **TIR** at the outside of the bend.

Key equations

$$\theta_c = \arcsin(n_{\text{clad}} / n_{\text{core}}) \cdot \text{sketch: } \theta_{i,\text{outer}} \approx \theta_{\text{ray}} + a/R$$

Frequently asked questions

Why does bending a fiber cause light to leak out? Doesn't the light still hit the interface?

Bending changes the geometry. In a straight fiber, a trapped ray hits the core-cladding interface at a steep angle (relative to the local normal). In a bend, especially on the outer wall of the curve, the ray approaches the interface at a shallower angle. If this new angle of incidence becomes smaller than the critical angle, TIR fails, and the ray refracts into the cladding, leading

to loss.

Does this model apply to all light in a real optical fiber?

This simulator is a simplified 2D model considering only 'meridional' rays that pass through the fiber axis. Real fibers also carry 'skew rays' that spiral without crossing the axis, which can have different bend loss characteristics. Furthermore, a full wave optics treatment considers guided modes, but this ray optics approach provides an intuitive and correct explanation for the primary mechanism of macrobend loss.

What is the real-world importance of understanding bend loss?

Bend loss sets practical limits on how tightly fiber optic cables can be coiled or routed in installations, such as in data centers, undersea cables, or fiber-to-the-home networks. Excessive bending causes signal attenuation and data errors. Conversely, specially designed 'bend-insensitive' fibers use advanced refractive index profiles to minimize this effect, allowing tighter bends in confined spaces.

How is the 'acceptance angle' related to bend loss?

The acceptance angle defines the maximum input angle for light to be initially guided in a straight fiber. A ray launched at this maximum angle is already at the critical angle for TIR. When such a marginally guided ray enters a bend, its effective incidence angle is most easily pushed below the critical angle, causing it to leak. Rays launched well within the acceptance cone are more robust to bending.

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NA from n_{core} , n_{clad} ; acceptance angle and bend loss cue.

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